

MATHEMATICAL FOUNDATIONS OF POLITICAL METHODOLOGY: PLSC 43401

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Course description

This is a first course on the theory and practice of mathematical methods in social science research. These mathematical and computer skills are needed for the quantitative and formal modeling courses offered in the political science department, and are increasingly necessary for courses in American Politics, Comparative Politics, and International Relations. We will cover mathematical techniques (linear algebra, calculus, probability) and methods of logical and statistical inference (proofs and statistics). A weekly computing lab will apply these methods, as well as introduce the R statistical computing environment.

Grading: You will be graded on weekly problem sets (worth 20% of your total grade), several programming assignments in R (also 20%), and two take-home exams (each worth 30% of your grade).

Problem sets are due each Tuesday at midnight, starting January 10. Upload your problem sets to Chalk.

Required books: *Mathematics for Economists*, 4th Edition, by Pemberton & Rau, and *Introduction to Probability*, 2nd Edition, by Bertsekas & Tsitsiklis.

JAN 3: INTRODUCTION: ALGEBRA, SETS, FUNCTIONS, AND PROOFS

- Pemberton & Rau Ch. 1-4
- **Week's R Goal:** Intro to R, data frames, Boolean logic, indexing, loops, functions.
- **R application:** NYC Flights

JAN 5: SEQUENCES, SERIES, LIMITS & CONTINUITY

- Pemberton & Rau Ch. 5

JAN 10: THE DERIVATIVE

- Pemberton & Rau Ch. 6
- **Week's R Goal:** Vectorized code with dplyr, joins.
- **R application:** Fearon Laitin and Religion

NO CLASS JAN 12: SPSA MEETINGS IN NEW ORLEANS

JAN 17: METHODS OF DIFFERENTIATION

- Pemberton & Rau Ch. 7
- **Week's R Goal:** Graphical summary of data, histograms, bar plots, group_by.
- **R application:** Language and geography of Afghanistan's villages

JAN 19: OPTIMIZATION

- Pemberton & Rau Ch. 8

JAN 24: INTEGRAL CALCULUS

- Pemberton & Rau Ch. 19
- **Week's R Goal:** tidyr
- **R application:** Police shootings

JAN 26: LINEAR ALGEBRA

- Pemberton & Rau Ch. 11

JAN 31: SYSTEMS OF LINEAR EQUATIONS & DETERMINANTS

- Pemberton & Rau Ch. 12, 13.1
- **Week's R Goal:** Purrr, advanced functions on lists

FEB 2: MULTIVARIATE CALCULUS & CONSTRAINED OPTIMIZATION

- Pemberton & Rau Ch. 14, 16, 17

Take-home Exam 1, due date TBA

FEB 7: LAWS AND PROPERTIES OF PROBABILITY

- Bertsekas-Tsitsiklis Ch. 1.1-1.2
- **Week's R Goal:** xtable, stargazer, reproducible code
- **R Application:** Data vs model of World Series lengths

FEB 9: CONDITIONAL PROBABILITY

- Bertsekas-Tsitsiklis Ch. 1.3-1.7

FEB 14: DISCRETE RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 2
- **Week's R Goal:** stringr, and other text tools
- **R Application:** Open Data Chicago: the Wage Gap

FEB 16: CONTINUOUS RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 3.1-3.3

FEB 21: SUMMARIZING RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 3.4-3.7
- **Week's R Goal:** ggmap and GIS

FEB 23: COVARIANCE AND CORRELATION

- Bertsekas-Tsitsiklis Ch. 4.1-4.3

FEB 28: TRANSFORMATIONS OF RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 4.4-4.4.6
- **Week's R Goal:** validate
- **R Application:** China's response to the TPP - a survey experiment

MAR 2: INEQUALITIES AND LIMITS OF SEQUENCES OF RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 5

MAR 7: HYPOTHESIS TESTING

- Bertsekas-Tsitsiklis Ch. 9
- Degroot and Schervish Chapter 8

Take-home Exam 2, due date TBA